4 **(a)** State the *principle of superposition*.

.....

.....

..... [2]

(b) Fig. 4.1 shows the variation with time t of the displacements x_A and x_B at a point P of two sound waves A and B.

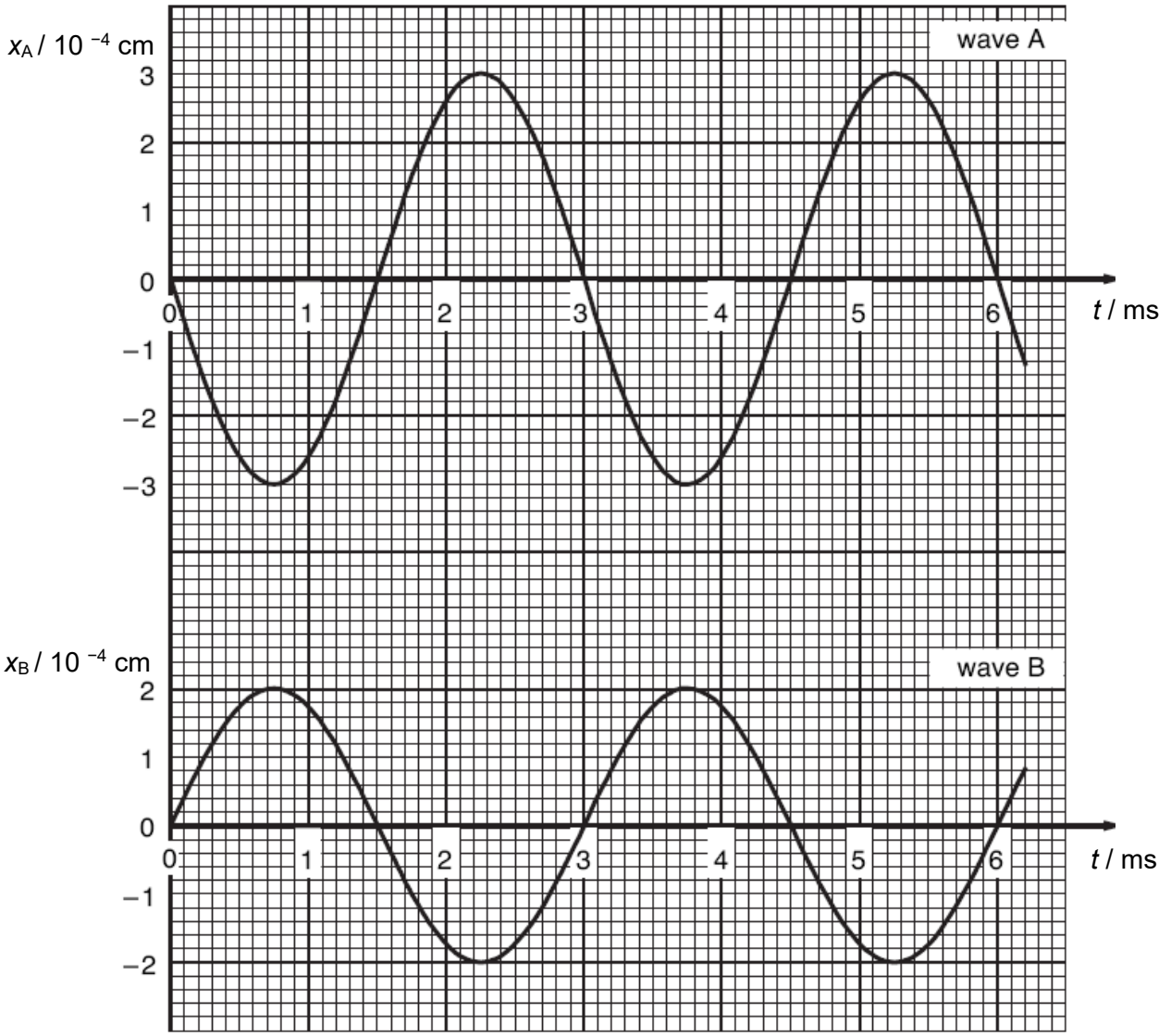

Fig. 4.1

(i) Determine the resultant displacement for the two waves at point P at time t = 4.0 ms,

resultant displacement = ...cm [1]

(ii) The intensity of wave A alone at point P is I. Calculate the resultant intensity, in terms of I, of the two waves at point P.

resultant intensity = ... [2]

(c) Fig. 4.2 shows the two loudspeakers that produced the two sound waves A and B as mentioned in **(b)**. Point O is equal distance from both loudspeakers.

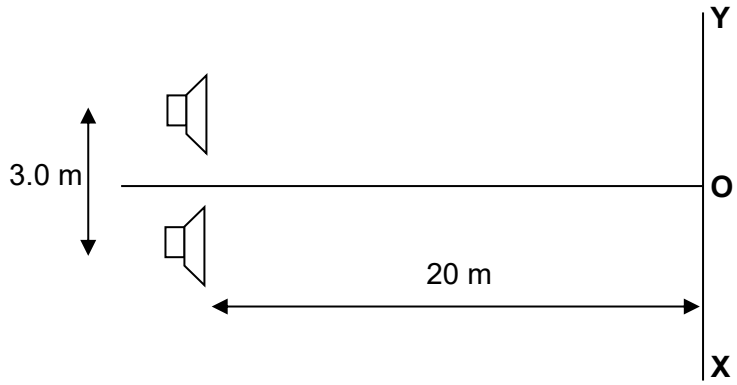

Fig. 4.2

(i) A detector is moved from point O towards Y along the line XY. After detecting three maxima including point O, a third minima is located at point P, which is 30.2 m from point O. Determine the speed of sound in air.

speed of sound = ... m s^{-1} [3]

(ii) Fig. 4.3 shows the two loudspeakers rotated to face each other but still kept 3.0 m apart. The sound emitted by the loudspeakers are in anti-phase.

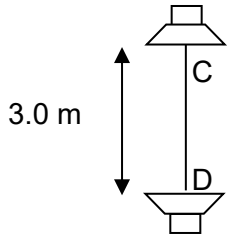

Fig. 4.3

If the detector is moved along the line CD, describe the variation of the sound intensity detected.

.....

..... [1]